

# Perturbations: gamma gates rates not just mean inhibition

exp042 · 2 June 2026

## Abstract

exp025's rate gap admits a cheap reading: PING fires less because the I-loop delivers more inhibition. This entry forecloses that reading and identifies *what* about the I-stream is doing the suppressing: qualitatively (rhythm vs mean), and quantitatively (which temporal precision is required). Scaffolded by ar009 §Leg 1 item 2.

## Methods

Pure inference on the trained exp025 PING baseline (seed 42,  $\theta_u = \text{off}$ ). For each batch the I-population spike tensor  $\mathbf{s}_{\text{base}}^I \in \{0, 1\}^{T \times B \times N_I}$  is recorded from a baseline forward pass, then an override tensor replaces it in a second pass via the exp037 hidden-perturbation hook. The E-population experiences only the override I-stream through  $W^{IE}$ ; the readout consumes the perturbed E spikes. Mean per-cell I rate is matched to the baseline to four decimals across every perturbation.

Five perturbation families:

1. **Baseline**: no override; trained PING dynamics.
2. **Cycle-coherent jitter**: partition the trial into blocks of length  $1/f_\gamma$  ( $\approx 22.8$  ms at the trained operating point from exp041). For each (trial, block), draw a single Gaussian offset  $\Delta \sim \mathcal{N}(0, \sigma^2)$  and shift every I-spike in that block by  $\Delta$ . Within-burst cross-cell synchrony is preserved exactly; only the *placement* of each burst is perturbed. Sweep  $\sigma \in \{0, 1, 3, 7, 14, 21, 28, 42, 60, 100\}$  ms.
3. **Per-I-cell jitter**: draw an independent Gaussian offset  $\Delta_{b,n,k} \sim \mathcal{N}(0, \sigma^2)$  for *every* I-spike and shift each spike by its own offset. Destroys within-burst cross-cell synchrony while keeping the mean per-cell rate; the within-burst counterpart of cycle-coherent jitter. Sweep  $\sigma \in \{0, 0.5, 1, 2, 5, 9, 14, 21, 50\}$  ms.
4. **Phase-shuffle**: per-trial permutation  $\pi_b$  of the time axis applied to all I-cells together:  $\mathbf{s}_{\text{shuf}[t,b,n]}^I = \mathbf{s}_{\text{base}[\pi_b(t),b,n]}^I$ . Preserves cross-cell co-firing within a timestep; destroys all phase structure.
5. **Rate-matched Poisson**: per-(trial, cell) Bernoulli with  $p = \text{count}_{b,n}/T$ . Destroys both temporal and cross-cell structure; tests the  $g_i$  variance limit.

## Results

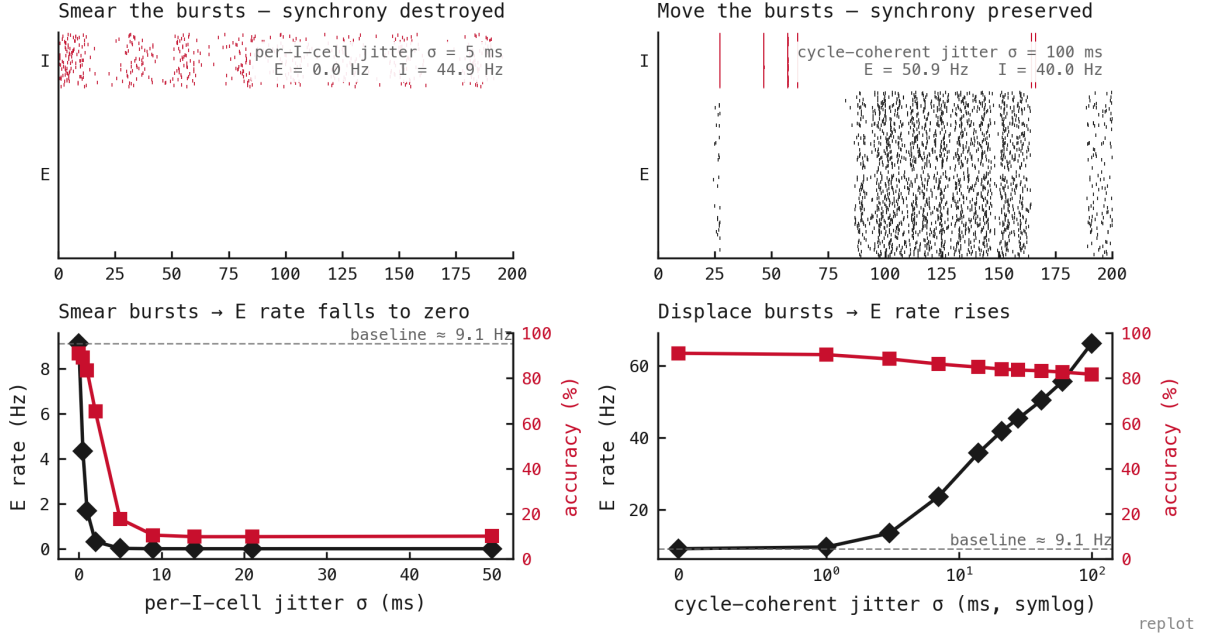
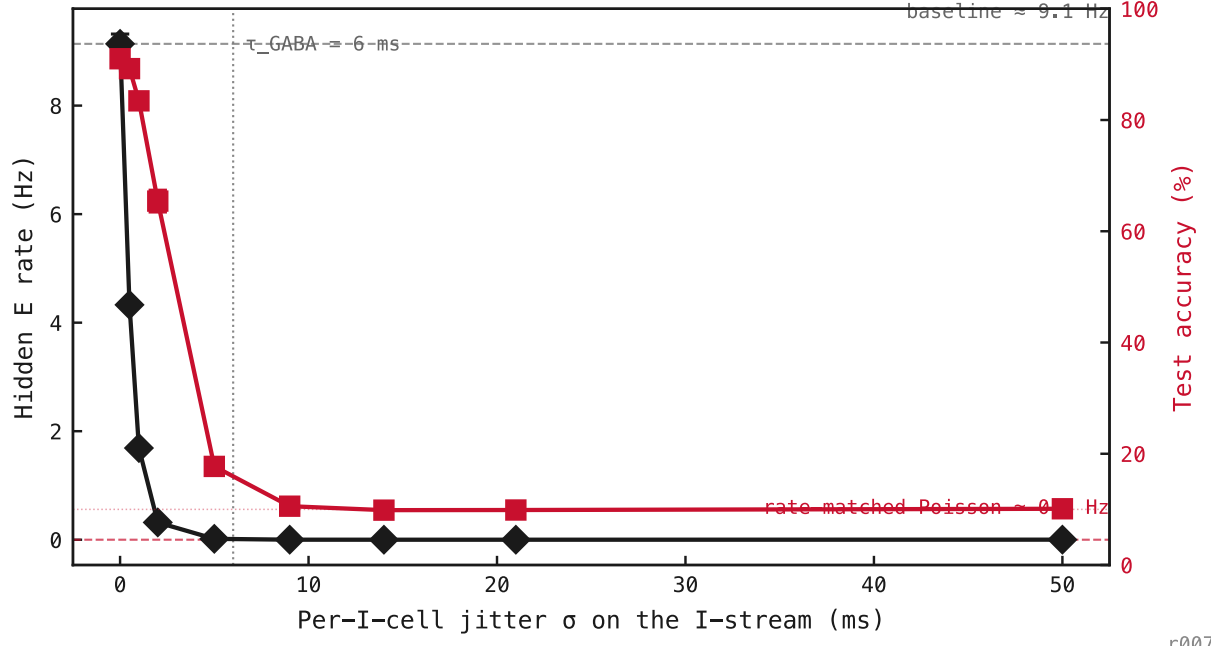


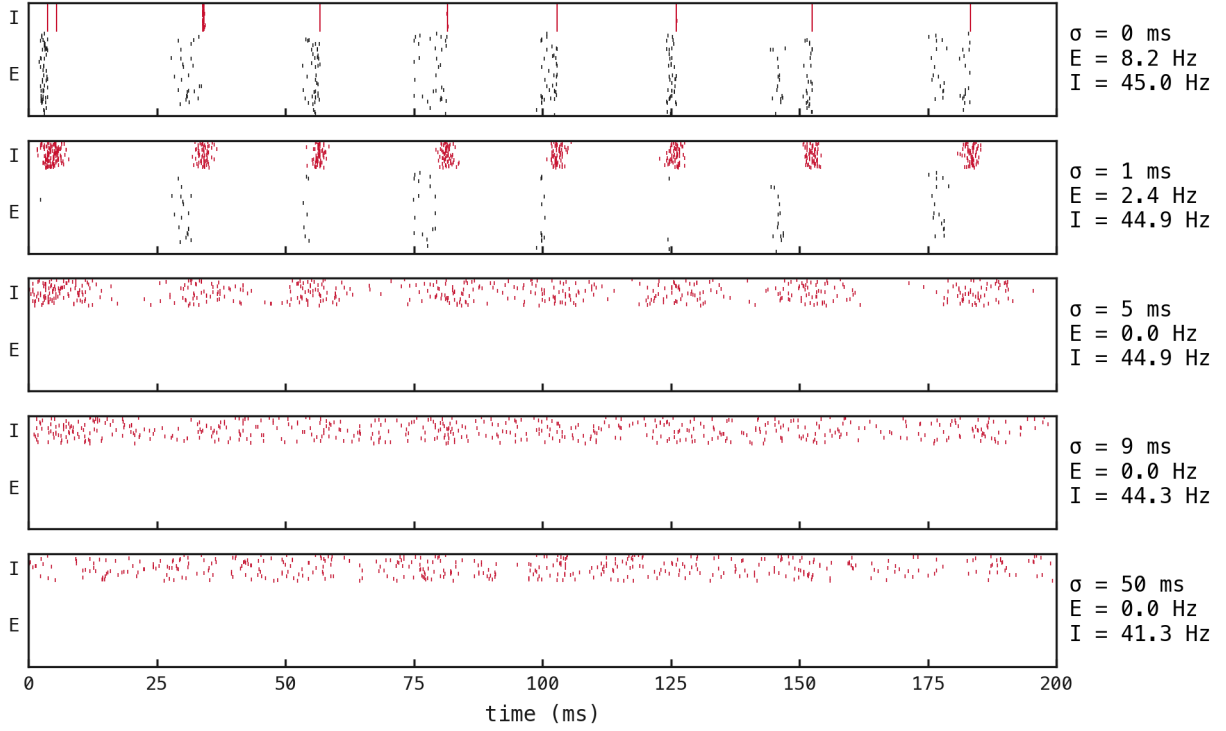
Figure 1: Two inference-time perturbations of the trained-PING I-stream, **both holding mean per-cell I rate fixed**, push the E rate in *opposite* directions, which a mean-inhibition account cannot produce. **Left column, smear the bursts:** per-I-cell jitter ( $\sigma = 5$  ms shown) scatters the spikes within each burst, destroying synchrony while leaving the mean untouched; the burst dissolves into a continuous shunt, the E rate falls to zero, and accuracy collapses toward chance (9.9% at the Poisson limit). **Right column, move the bursts:** cycle-coherent jitter ( $\sigma = 100$  ms shown) displaces each gamma burst bodily but keeps its within-burst synchrony; the I-stream opens gaps and the E rate *rises* from 9.1 Hz at baseline to 66.3 Hz, accuracy holding near 81.7%. Same mean inhibition, opposite outcome: what gates the E rate is the *timing* of inhibition, the rhythm, not its average level.

## Per-I-cell jitter



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Figure 2: Per-I-cell jitter sweep, three seeds. Each spike receives an independent Gaussian offset; mean per-cell I rate is preserved exactly. **E rate (black diamonds, left axis) falls monotonically from baseline (9.1 Hz)**, already more than halved to 4.3 Hz by  $\sigma = 0.5$  ms and essentially zero (0 Hz) by  $\sigma \approx 5$  ms, below  $\tau_{\text{GABA}} = 6$  ms. **Accuracy (red squares, right axis) holds at  $\approx 89.2\%$  up to  $\sigma = 0.5$  ms**, then collapses through 83.4% ( $\sigma = 1$ ), 65.4% ( $\sigma = 2$ ) and 17.7% ( $\sigma = 5$ ), bottoming at chance (10.6%) by  $\sigma = 9$  ms. The  $\sigma \rightarrow \infty$  asymptote is the rate-matched Poisson regime: E silent, accuracy at chance.



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Figure 3: Single trial replayed at five per-cell jitter levels. At  $\sigma = 0$  the I-bursts are crisp vertical bands. **At  $\sigma = 1$  ms** the bursts visibly smear into a few-ms-wide cluster and E firing already collapses to the low single digits (per-trial E annotated on each panel; sweep mean 1.7 Hz). **At  $\sigma \geq 5$  ms** the I-stream looks indistinguishable from a continuous low-variance shunt, and E is silenced. Per-cell jitter doesn't release E; it destroys the bursty structure that gave E its recovery troughs in the first place.

### Cycle-coherent jitter

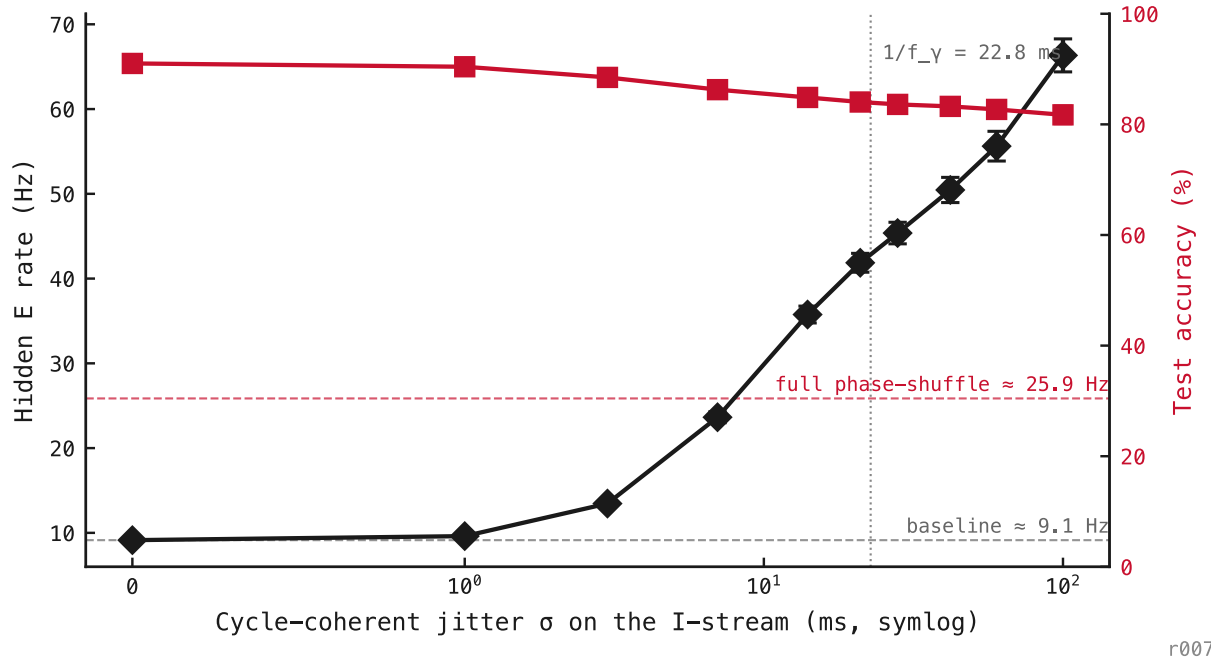


Figure 4: E rate (black diamonds) and accuracy (red squares) vs cycle-coherent jitter  $\sigma$ , three seeds. Horizontal dashed lines mark the baseline rate (9.1 Hz) and the full phase-shuffle level (25.9 Hz, the reference with within-burst structure destroyed); the vertical dotted line sits at  $\sigma = 1/f_\gamma \approx 22.8$  ms, the predicted transition timescale. As  $\sigma$  grows the displaced bursts open wider gaps and the E rate climbs *past* the phase-shuffle level to 66.3 Hz by  $\sigma = 100$  ms, while accuracy declines only gently, holding near 81.7%.

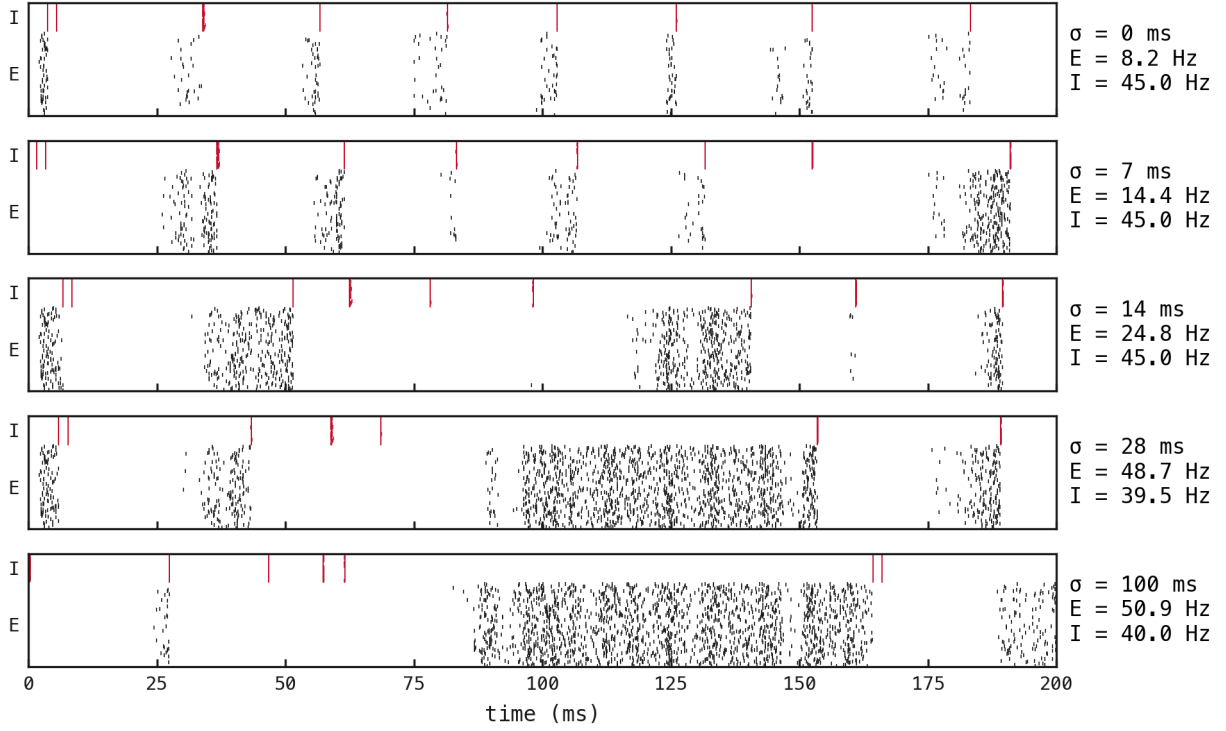


Figure 5: Single trial replayed at five jitter levels ( $\sigma = 0, 7, 14, 28, 100$  ms; seed 42, MNIST digit 0 sample 0). Per-trial E rate annotated on each panel. **The I-bands stay vertical and crisp at every  $\sigma$ :** within-burst synchrony is preserved exactly. What changes is *where* each burst lands: at larger  $\sigma$  the bursts are displaced bodily from their phase-locked positions, opening longer gaps in the I-stream that E fires through, and the E rate climbs accordingly.